Jia-Shu Pan 潘嘉书

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Research Focus

I study the **interplay between representation learning and generative modeling**:

1. How does the **encoding distribution** $q(z_0 \mid x)$ shape the **learnability of generative models**?
2. How does the **forward process** $p_F(z_{0:T})$ shape the **learnability of generative models**?
3. How can **representation-aware objectives** enable **few-step or one-step generation**?

These questions guide my current focus on **end-to-end (single-step) latent generative models**: jointly learning representations and generators to achieve both **efficient learning and few-step generation**.

Education and Research Visit

2025 - present	Ph.D. Student *School of Engineering, Westlake University* Advisor: Prof. Tailin Wu
2023 - 2025	Research Visit *Australian National University* Worked with Prof. Yuan-Sen Ting
2018 - 2023	Undergraduate, Astronomy *Nanjing University*

Research Highlight

Feature Information Dynamics in Diffusion *NeurIPS 2026*

Diffusion models are often said to generate coarse structure before fine detail. We asked whether this intuition still holds when data are mapped into different representation spaces.

- Using the I-MMSE relation, we defined **feature information density** to measure when class, mask, and Canny information becomes accessible during denoising.
- We found that **representation reshapes the order of feature emergence** across pixel, SDVAE, VAVAE, and RAE spaces. Only RAE follows the class, then mask, then Canny order.
- Diffusion models also converge fastest in RAE among these spaces. We hypothesize that its **ordered feature dynamics** helps explain this difference.

Selected Publications

Feature Information Dynamics in Diffusion **NeurIPS 2026**
Jia-Shu Pan, Tao Zhang, Yufei Huang, Yanjun Sheng, Tailin Wu.

VFScale: Intrinsic Reasoning through Verifier-Free Test-time Scalable Diffusion Model **ICLR 2026**
Tao Zhang*, **Jia-Shu Pan***, Ruiqi Feng, Tailin Wu.
* Equal contribution.
Uses the diffusion model's intrinsic energy as a verifier, together with hybrid Monte Carlo Tree Search, to scale reasoning at test time.

Referees

Prof. Tailin Wu Westlake University	Assistant Professor *wutailin@westlake.edu.cn*
Prof. Yuan-Sen Ting Ohio State University	Associate Professor *ting.74@osu.edu*

CV design adapted from Prof. Yuan-Sen Ting's curriculum vitae.